

# Curried Functional Spaces

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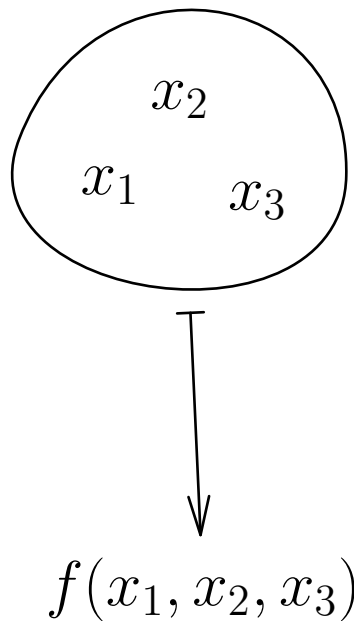
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May 13, 2026

# Premise

## Multivariable functions

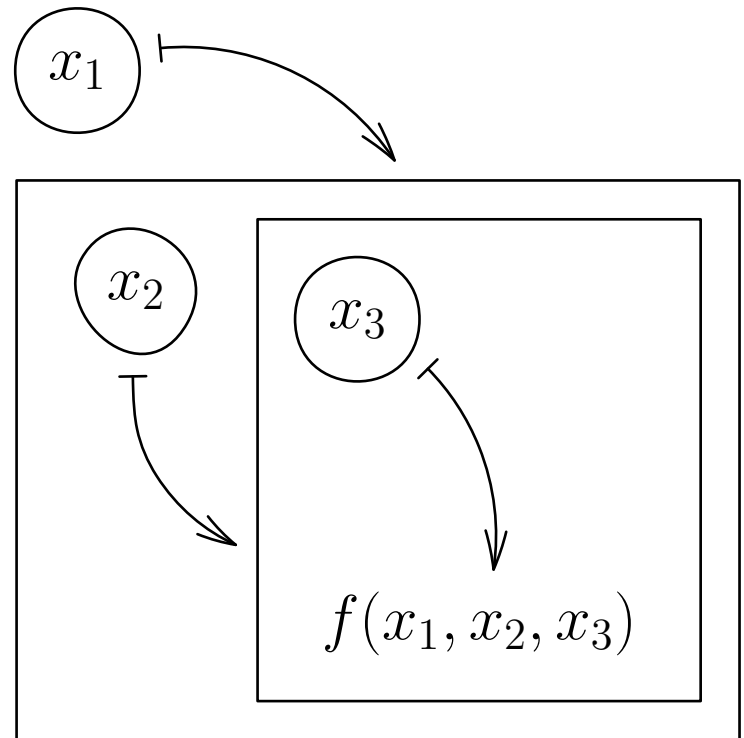
$$\mathbb{R}^n \rightarrow \mathbb{R}$$



## Multivariate calculus

## Curried functions

$$\mathbb{R} \rightarrow (\mathbb{R} \rightarrow (\dots (\mathbb{R} \rightarrow \mathbb{R})))$$

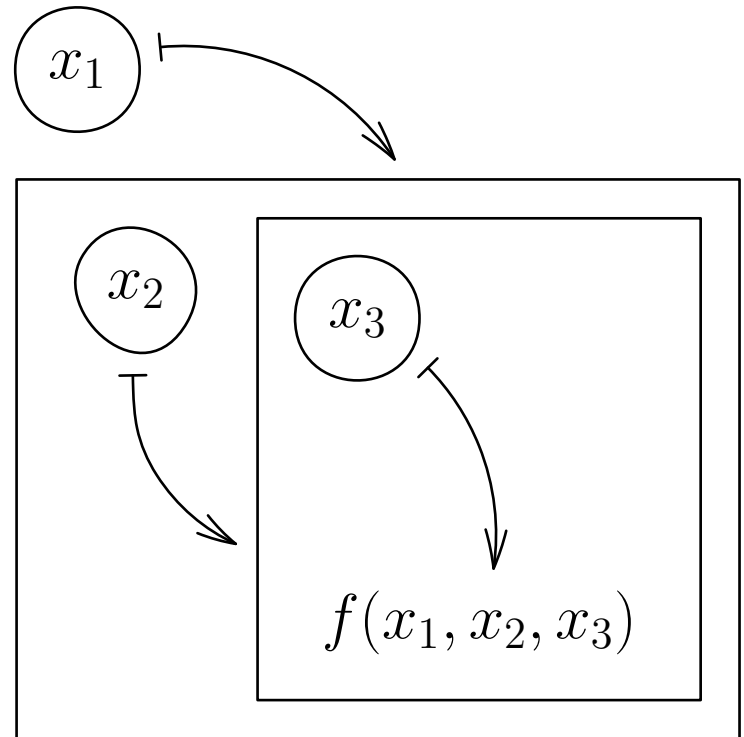


## $\lambda$ -calculus, functional programming

# Premise

## Curried functions

$$\mathbb{R} \rightarrow (\mathbb{R} \rightarrow (\dots (\mathbb{R} \rightarrow \mathbb{R})))$$



$\lambda$ -calculus,  
functional programming

Multivariate calculus

# Continuous Functions

$C_n(\mathbb{R})$  — the space of continuous  $n$ -ary curried functions:

1.  $C_0(\mathbb{R}) := \mathbb{R}$

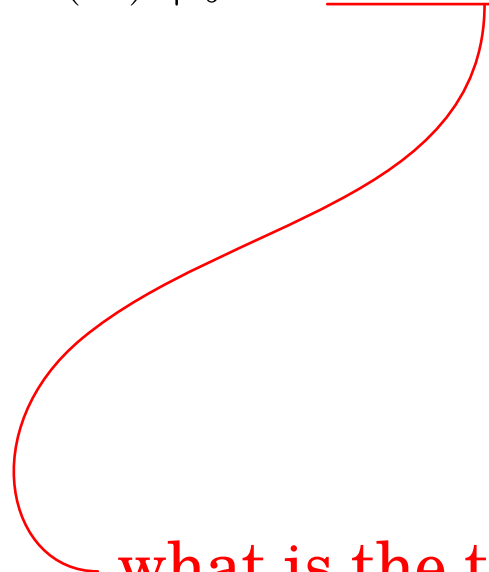
2.  $C_{n+1}(\mathbb{R}) := \{f: \mathbb{R} \rightarrow C_n(\mathbb{R}) \mid f \text{ is continuous}\}$

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what is the topology on  $C_n(\mathbb{R})$ ?

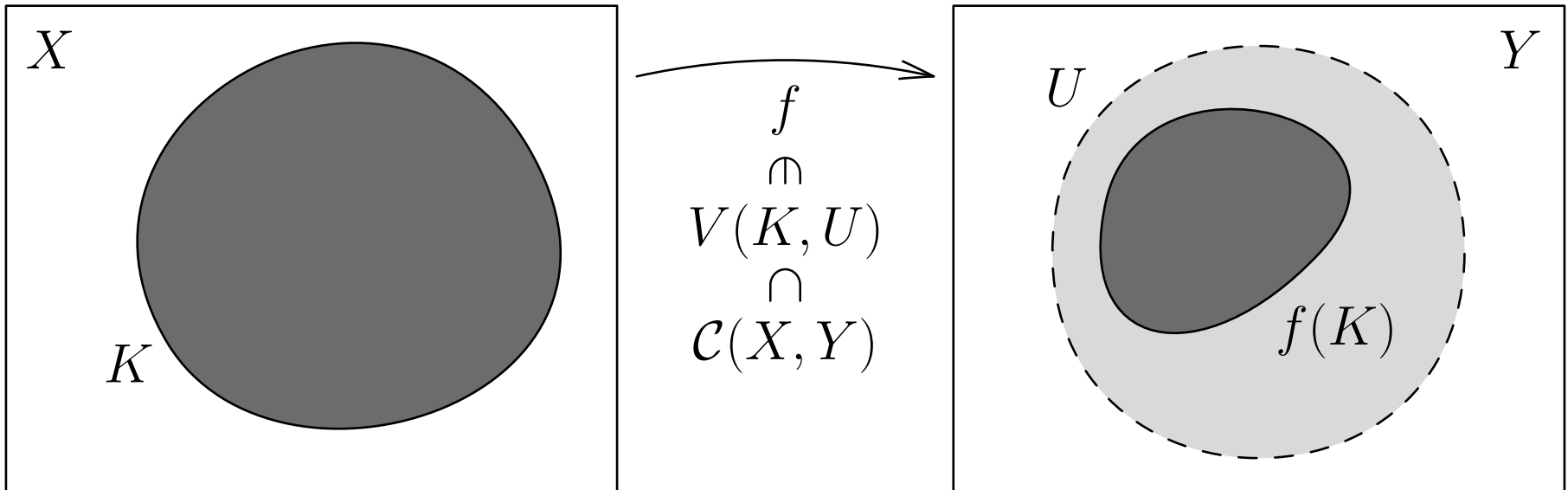
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compact-open topology:  $C_n(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R}))$



# Topological Vector Spaces

**Remark:** If  $f_1, f_2 \in C_n(\mathbb{R})$ , then  $\alpha f_1 + \beta f_2 \in C_n(\mathbb{R})$ .

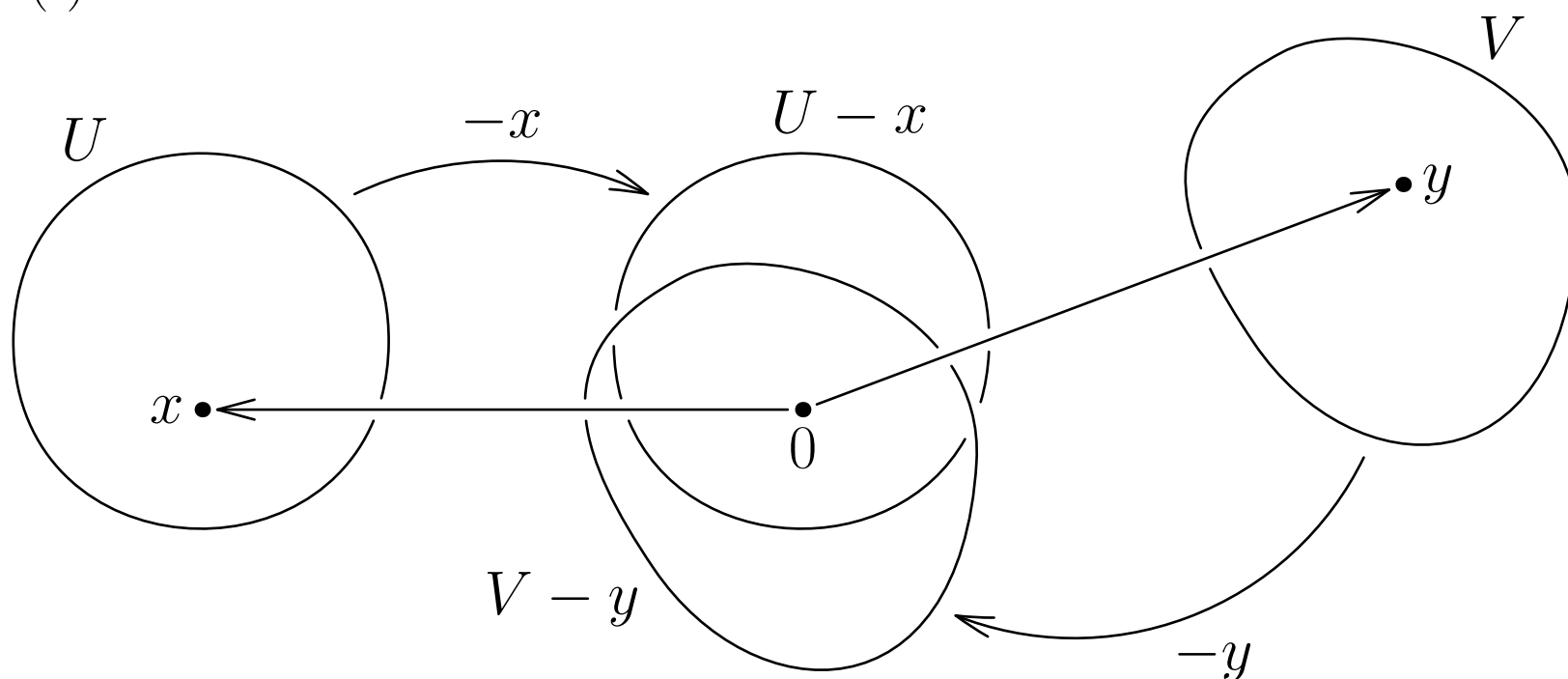
# Topological Vector Spaces

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**Definition:**  $X$  is a **TVS**  $\Leftrightarrow X$  has **topological** and **vector** structure, and they are *compatible*:

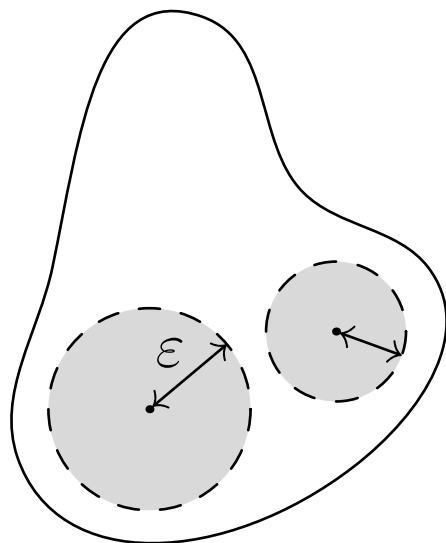
1.  $(+): X \times X \rightarrow X$  is continuous;
2.  $(\cdot): \mathbb{R} \times X \rightarrow X$  is continuous.



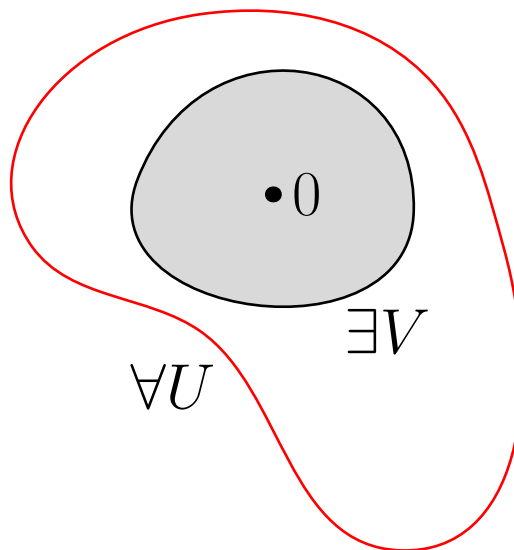


# Properties of Topological Vector Spaces

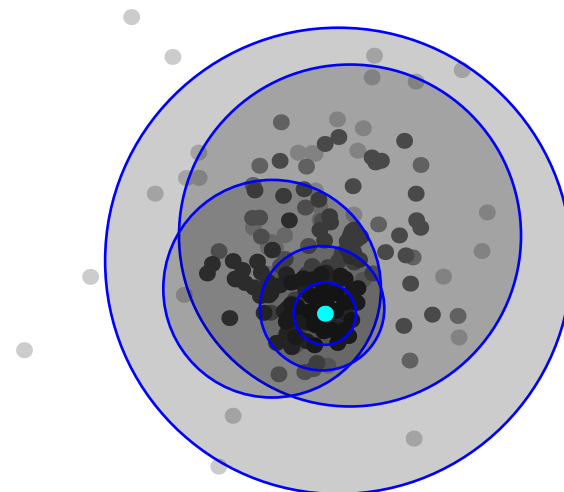
Metrizability



Local convexity



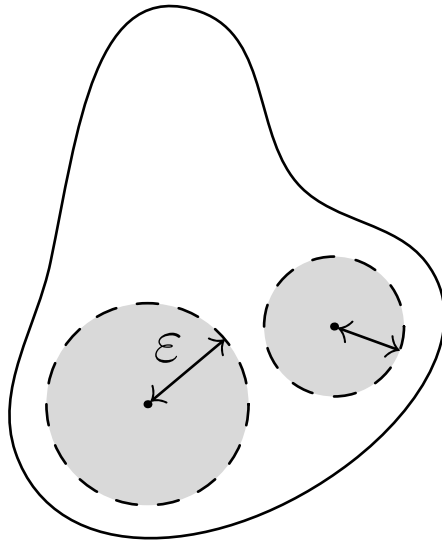
Completeness



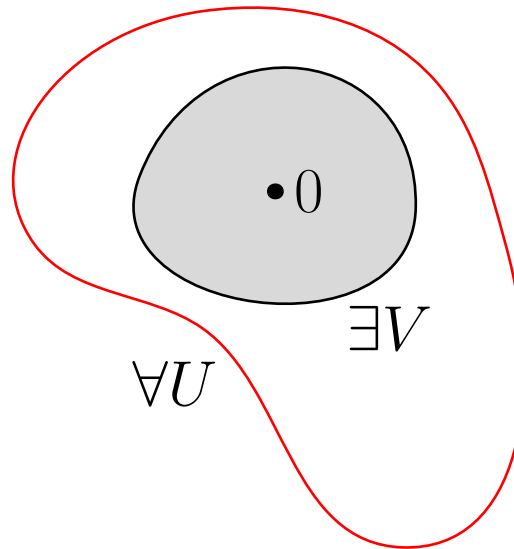
Fréchet space

# Properties of Topological Vector Spaces

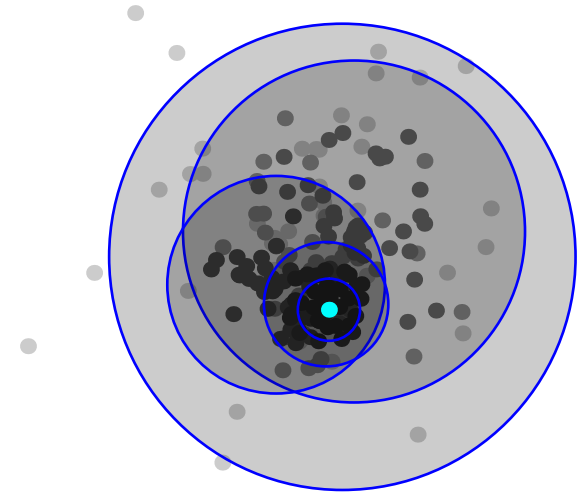
Metrizability



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Completeness



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Curried function spaces  $C_n(\mathbb{R})$

# Continuous Functions

$C_n(\mathbb{R})$  — the space of continuous  $n$ -ary curried functions:

1.  $C_0(\mathbb{R}) := \mathbb{R}$ ;
  2.  $C_{n+1}(\mathbb{R}) := \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$ , with the compact-open topology.
- 

**Results:** for all  $n \in \mathbb{N}$ ,

1.  $C_n(\mathbb{R})$  is a Hausdorff topological vector space;
  2. There is an isomorphism  $\gamma: \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \leftrightarrow C_n(\mathbb{R})$ ;
  3.  $C_n(\mathbb{R})$  is a Fréchet space.
- 

$\Rightarrow$  Theorems from functional analysis apply:

1. Uniform boundedness principle;
2. Open mapping theorem;
3. ...

# Differentiable functions

$D_n^m(\mathbb{R})$  — the space of  $n$ -ary  $m$  times differentiable functions:

1.  $D_0^m(\mathbb{R}) := \mathbb{R};$

2.  $D_n^0(\mathbb{R}) := C_n(\mathbb{R});$

3. If  $n, m > 0$ , then  $D_n^m(\mathbb{R})$  consists of all such  $f \in \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R}))$ , that there is  $f' \in D_n^{m-1}(\mathbb{R})$  with

$$\frac{f(x+h) - f(x)}{h} \xrightarrow{h \rightarrow 0} f'(x).$$

# Differentiable functions

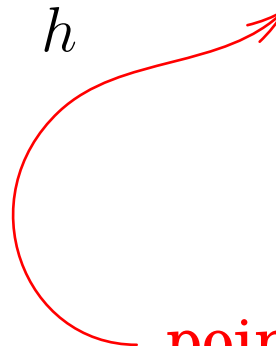
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pointwise convergence!

$$f_k \xrightarrow{k \rightarrow \infty} f \Leftrightarrow f_k(x_1) \dots (x_n) \xrightarrow{k \rightarrow \infty} f(x_1) \dots (x_n)$$

# Differentiable functions

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The topology on  $D_n^m$  is *coarsest* such that the maps

$$\begin{array}{ccc} i: D_n^m(\mathbb{R}) \rightarrow \mathcal{C}(\mathbb{R}, D_{n-1}^m(\mathbb{R})), & d: D_n^m(\mathbb{R}) \rightarrow D_n^{m-1}(\mathbb{R}) \\ f \mapsto f & f \mapsto f' \end{array}$$

are continuous. In particular,  $f_k \rightarrow f \Rightarrow f'_k \rightarrow f'$ .

# Differentiable functions

$D_n^m(\mathbb{R})$  — the space of  $n$ -ary  $m$  times differentiable functions:

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**Results:** for all  $m, n \in \mathbb{N}$ ,

1.  $D_n^m(\mathbb{R})$  is a Hausdorff topological vector space;
2. The inclusion map  $j: D_n^{m+1}(\mathbb{R}) \rightarrow D_n^m(\mathbb{R})$  is continuous;
3. There is an isomorphism  $\gamma: C^m(\mathbb{R}^n) \leftrightarrow D_n^m(\mathbb{R})$ , given by

$$\gamma\left(\tilde{f}\right)(x_1)(x_2)\dots(x_n) = \tilde{f}(x_1, x_2, \dots, x_n).$$

4.  $D_n^m(\mathbb{R})$  is a Fréchet space.

# Future directions

1. Apply results from functional analysis to derive facts about  $C_n(\mathbb{R})$  and  $D_n^m(\mathbb{R})$ ;
2. Study complex curried functions,  $C_n(\mathbb{C})$  and  $D_n^m(\mathbb{C})$ ;
3. Define Riemann integrals on  $C_n(\mathbb{R})$ ;
4. Prove analogs of the Inverse/Implicit function theorems.



# References

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- [2] N. Bourbaki, *General Topology, Part II*. 1966
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- [4] W. Rudin, *Principles of Mathematical Analysis*. 1976
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